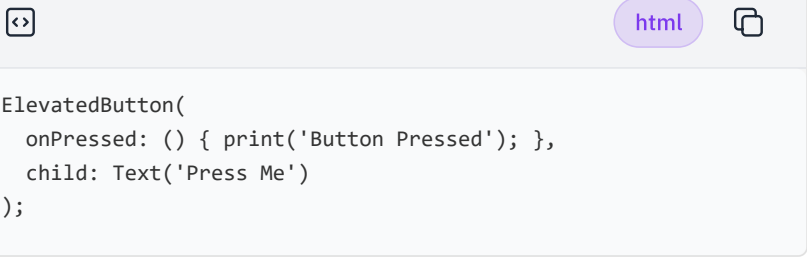
Cheat Sheet: Working with Flutter

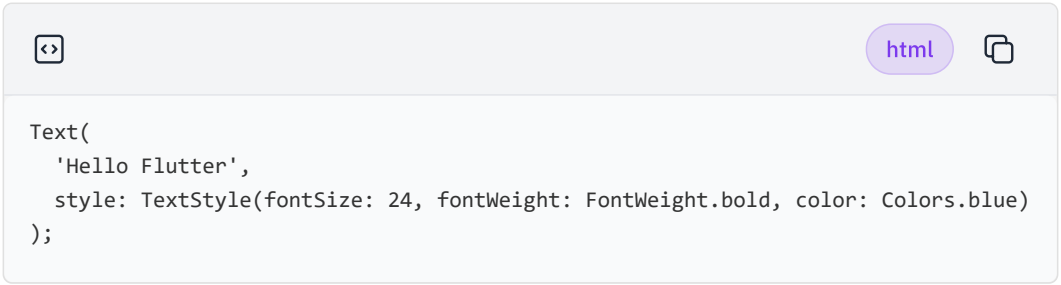
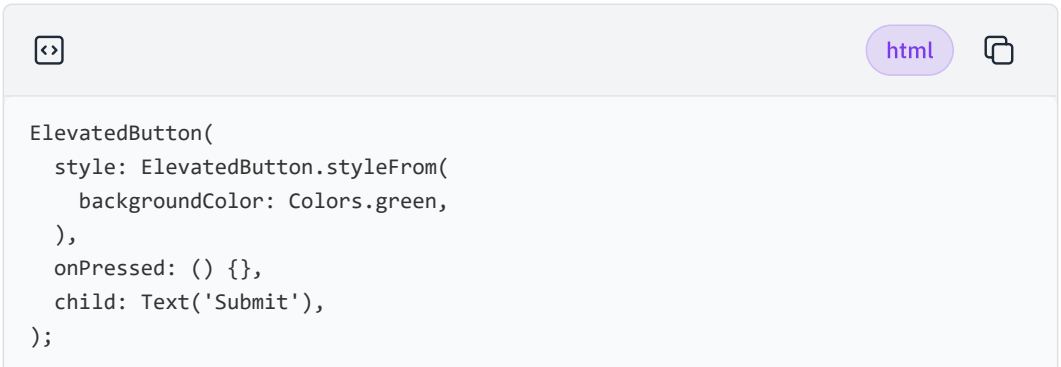
Interaction and forms in Flutter

Type	Description	Code example
ElevatedButton	Triggers actions on user interaction	
TextField	Collects textual input from users	
Form validation	Ensures input meets certain criteria	
GestureDetector	Handles more complex gestures like swipes	

Routing in Flutter

Type	Description	Code example
Basic navigation	Manages screen transitions within an app	
Named routes	Uses named routes for navigation, which simplifies route management	
Passing data between routes	Allows data to be sent between screens	

Implementing styles in Flutter

Type	Description	Code example
Text widget styling	Customizes the appearance of text elements	
Styling buttons	Modifies the look of buttons to align with the app design	

Navigation in Flutter

Type	Description	Code example
Stack navigation	Manages a stack of screens	

```
Navigator.push(
  context,
  MaterialPageRoute(builder: (context) => SecondScreen())
);
```

Tab navigation	Organizes content across different tabs	

```
TabBar(
  tabs: [Tab(icon: Icon(Icons.directions_car)), Tab(icon: Icon(Icons.directions_transit))]
);
```

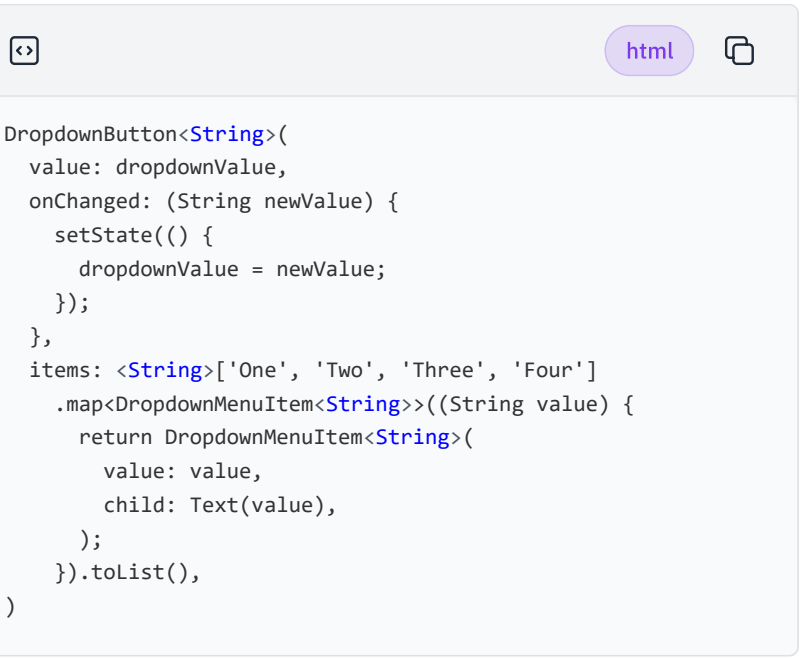
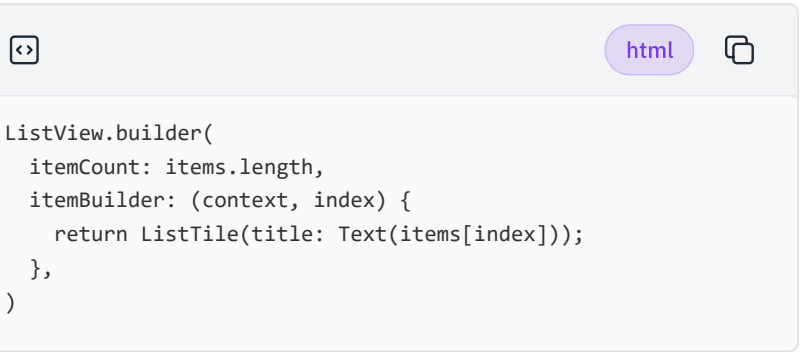
Bottom navigation bar	Provides navigation between top-level views of an app through a series of tabs at the bottom of the screen	

```
BottomNavigationBar(
  items: <BottomNavigationBarItem>[
    BottomNavigationBarItem(icon: Icon(Icons.home), label: 'Home'),
    BottomNavigationBarItem(icon: Icon(Icons.business), label: 'Business'),
  ],
  onTap: (index) { /* Handle tap */ },
);
```

Drawer navigation	Offers a slide-in menu from the edge of the screen for navigation purposes	

```
Drawer(
  child: ListView(
    children: <Widget>[
      DrawerHeader(child: Text('Header')),
      ListTile(title: Text('Item 1'), onTap: () {}),
    ],
  ),
);
```

Flutter widgets

Type	Description	Code example
DropdownButton	Allows users to select from a list of items in a dropdown menu	
Flexible widget	Allows child widgets to flex within available space in a row or column	
ListView.builder	Constructs a list view dynamically with an item builder function	
SnackBar	Provides lightweight feedback about an operation by showing a brief message	

Dialogs

Displays material design dialogs

```
showDialog(
  context: context,
  builder: (BuildContext context) {
    return AlertDialog(
      title: Text('Alert!'),
      content: Text('You have been alerted.'),
      actions: <Widget>[
        TextButton(
          onPressed: () {
            Navigator.of(context).pop();
          },
          child: Text('Close'),
        ),
      ],
    );
  },
);
```

Padding widget

Applies padding to its child widget

```
Padding(
  padding: EdgeInsets.all(8.0),
  child: Text('Hello World!'),
)
```

GridView

Creates a scrollable grid of widgets arranged in rows and columns

```
GridView.count(
  crossAxisCount: 2,
  children: List.generate(20, (index) {
    return Center(
      child: Text('Item $index'),
    );
  }),
)
```

ExpansionPanel	Provides a way to expand and collapse content	 html 
		`ExpansionPanelList(`
		`  expansionCallback: (int index, bool isExpanded) {`
		`  setState(() {`
		`  items[index].isExpanded = !isExpanded;`
		`  });`
		`  },`
		`  children: items.map<ExpansionPanel>((Item item) {`
		`  return ExpansionPanel(`
		`  headerBuilder: (BuildContext context, bool isExpanded) {`
		`  return ListTile(title: Text(item.headerValue));`
		`  },`
		`  body: ListTile(title: Text(item.expandedValue)),`
		`  isExpanded: item.isExpanded,`
		`  );`
		`  }).toList(),`
		`)`

Custom paint	Provides a canvas on which to draw during the paint phase	 html 
		`CustomPaint(`
		`  painter: MyPainter(),`
		`)`

Theme data	Defines the color and typography values for a Material app	 html 
		`ThemeData(`
		`  colorScheme: ColorScheme.fromSeed(`
		`  seedColor: Colors.blue,`
		`  ),`
		`  useMaterial3: true,`
		`);`

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GestureDetector	Handles more complex gestures like swipes	

Routing in Flutter

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Basic navigation	Manages screen transitions within an app	html
		`Navigator.push(`
		`  context,`
		`  MaterialPageRoute(builder: (context) => NewScreen())`
		`);`

Named routes	Uses named routes for navigation, which simplifies route management	html `Navigator.pushNamed(context, '/details');`

Passing data between routes	Allows data to be sent between screens	html
		`Navigator.push(`
		`  context,`
		`  MaterialPageRoute(`
		`  builder: (context) => DetailScreen(data: 'Data from previous screen')`
		`  )`
		`);`

Implementing styles in Flutter

Type	Description	Code example

Text widget styling	Customizes the appearance of text elements	html
		`Text(`
		`  'Hello Flutter',`
		`  style: TextStyle(fontSize: 24, fontWeight: FontWeight.bold, color: Colors.blue)`
		`);`

Styling buttons	Modifies the look of buttons to align with the app design	html
		`ElevatedButton(`
		`  style: ElevatedButton.styleFrom(`
		`  backgroundColor: Colors.green,`
		`  ),`
		`  onPressed: () {},`
		`  child: Text('Submit'),`
		`);`

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Flutter widgets

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DropdownButton	Allows users to select from a list of items in a dropdown menu	html
		`DropdownButton<String>(`
		`  value: dropdownValue,`
		`  onChanged: (String newValue) {`
		`  setState(() {`
		`  dropdownValue = newValue;`
		`  });`
		`  },`
		`  items: <String>['One', 'Two', 'Three', 'Four']`
		`  .map<DropdownMenuItem<String>>((String value) {`
		`  return DropdownMenuItem<String>(`
		`  value: value,`
		`  child: Text(value),`
		`  );`
		`  }).toList(),`
		`)`

Flexible widget	Allows child widgets to flex within available space in a row or column	html
		`Row(children: [`
		`  Flexible(child: Container(color: Colors.red, height: 50)),`
		`  Flexible(child: Container(color: Colors.blue, height: 50))`
		`])`

ListView.builder	Constructs a list view dynamically with an item builder function	html
		`ListView.builder(`
		`  itemCount: items.length,`
		`  itemBuilder: (context, index) {`
		`  return ListTile(title: Text(items[index]));`
		`  },`
		`)`

SnackBar	Provides lightweight feedback about an operation by showing a brief message	html
		`final snackBar = SnackBar(content: Text('Yay! A SnackBar!'));`
		`ScaffoldMessenger.of(context).showSnackBar(snackBar);`

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  context: context,
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      content: Text('You have been alerted.'),
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